

Junghoon Chung

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Summary

PhD candidate in Industrial & Systems Engineering with a strong background in **UX research, human factors, and data-driven design**. Experienced in **designing and executing end-to-end mixed-methods research** including interviews, surveys, controlled experiments, and field studies. Skilled at **integrating qualitative insights with large-scale behavioral and sensor data** to inform product and system design. Proven ability to **translate complex findings into actionable recommendations for cross-functional teams**. Seeking a UX Research internship to support evidence-driven product development at scale.

Education

University of Washington, Seattle	Sep 2025 – Jun 2029 (Expected)
Ph.D in Industrial and Systems Engineering, Dean's Fellowship Recipient	
Virginia Tech, Blacksburg	Aug 2023 – May 2025
M.S in Industrial and Systems Engineering (Thesis Track), GPA 3.89/4.0	
Yonsei University, Seoul	Mar 2017 – Feb 2023
B.S in Industrial Engineering, GPA 3.52/4.0, Student Council President	

Experience

Graduate Research Assistant, Human Automation Systems Lab, University of Washington	Sep 2025 – Present
– Designed controlled experiments and user studies to evaluate attention, workload, and performance in complex task environments – Collected multimodal behavioral and physiological data to understand user states during sustained interaction – Generated insights for designing systems that support sustained attention and reduce cognitive overload	
Graduate Research Assistant, Occupational Ergonomics and Biomechanics Lab, Virginia Tech	Aug 2023 – May 2025
– Led end-to-end mixed-methods research on how knowledge workers interact with computer systems and workplace technologies – Conducted interviews, surveys, and behavioral observations alongside sensor-based data collection – Synthesized qualitative feedback and quantitative data to identify usability barriers and contextual drivers of behavior – Analyzed longitudinal data to understand patterns in productivity, comfort, and task engagement	

Key Projects

- **Human Vigilance Modeling with causal-inference models**, — University of Washington
Investigated how cognitive load and alertness influence performance during sustained tasks by integrating behavioral measures with **30+ physiological and contextual features across over 10,000 observations per participant**. Developed predictive models that **improved performance by 20%** through feature engineering and multimodal integration, generating insights for designing systems that better support attention and reduce cognitive burden in safety-critical environments.
- **Context-Aware Sit-Stand Intervention for Promoting Healthy Behaviors in Knowledge Workers**, — Virginia Tech
Conducted a longitudinal field study using **3 weeks of behavioral logs per participant (over 100,000 records)** to understand posture-switching behavior during computer work. **Applied time-series forecasting and user clustering to identify contextual triggers of behavior change**, improving prediction accuracy through segmentation of usage patterns and informing the design of adaptive workplace interventions that balance productivity and health.
- **Review reliability analysis by BERT sentiment analysis and Clustering** — Yonsei University
Analyzed over **50,000 online product reviews using transformer-based sentiment modeling and clustering** to identify bias patterns across reviewer groups. Proposed **cluster-normalized scoring methods to reduce user bias and improve the reliability** of aggregate product ratings derived from large-scale user-generated data.

Skills

Programming: Python, SQL (MySQL)
Research Methods: Interviews, Surveys, Usability Testing, Experimental Design
Analysis: Statistical analysis, Evidence synthesis
Collaboration: Cross-functional teamwork, Technical communication, Presentation

Awards

Dean's Fellowship, College of Engineering	University of Washington, 2025 –2029
Social-Innovation Capability Video Contest 2nd Place – Data mining	Yonsei University, 2022
Merit-based Scholarship (Academic Excellence)	Yonsei University, 2021